


# SOFTWARE DESIGN 2

## GROUP 32 AUTOVAULT USER GUIDE

**Group members**

| Member | Name | Student number |
|---|---|---|
| 1 | Mtshali Aphiwe | 250858908 |
| 2 | Mavhandu Khano | 251056864 |
| 3 | Magasha Sanelisiwe Luyema | 250237210 |
| 4 | Nemahate Orinea | 250799820 |
| 5 | Thukhutha Ampharisaho | 250086387 |
| 6 | Moswane Kelebogile | 230467083 |

# AutoVault: User Guide

AutoVault is a backup tool that copies chosen folders on a schedule, keeps a history of versions, and lets you restore an earlier version.

**Team:** Aphiwe Mtshali (Project Lead), Khano Mavhandu, Luyema Maqasha, Orinea Nemahate, Ampfarisaho Thukhutha, Kelebogile Moswane.

## What AutoVault Does

- Copies a folder (source) to a backup folder as a new version every time.
- Backs up only when there is something to back up – when a file has beenmodified, added or deleted.
- Ignores files that have certain extensions, such as .tmp or .log.
- Stores 5 newest versions of each job and removes other ones.
- Allows restoring a chosen version to a chosen folder.
- Can perform automatic backups at regular intervals when the software is working.
- Automatically saves your jobs and writes a history log.

## What you need

- A computer with Java installed. AutoVault was built and tested with Java 21.
- The AutoVault project (NetBeans project AutoVault), or the packaged file dist/AutoV.jar.

## Launching AutoVault

In NetBeans: open AutoVault project, set ui.Main as main class and press F6. AutoVault window launches and console menu opens simultaneously in Output window. Both are working on the same list of jobs..

## Working with window

The window presents a table of your jobs with columns Name, Source, Destination, Every (minutes), Last run and Versions. It updates every 5 seconds.

| Button | What it does |
|---|---|
| **Add job** | Opens a form. Fill in the name, the source folder, the backup folder, how often (in minutes), and the extensions to skip (separated by commas, for example tmp,log). Use **Browse...** to pick folders |
| **Backup now** | Backs up the job you selected in the table |
| **Versions / Restore** | Lists the versions of the selected job. Choose one, then |

| | |
|---|---|
| | choose the folder to restore into |
| **Remove job** | Removes the selected job after you confirm. The backup files already made are kept |
| **QA / Logging** | Opens a new window where you can view activity logs, run quality assurance tests, and export the log file. |
| **Save** | Saves your jobs now |

The line at the bottom of the screen will show you what was done, such as "Backup finished successfully" or "There were no changes since the last backup, there is nothing to do." You exit the program by closing the window.

### How Backup is Done

1.Initial backup: All files in the source folder are copied to a new version folder in the backup folder.
2.Following backups: AutoVault checks for changes first; if there were none, it outputs "No changes ... Nothing to do" and does not make a new version; otherwise, it creates a new version if a file is modified, added, or deleted.
3.A version is a complete copy of the source folder that can be restored individually.
4.Skipped extensions are never copied and changes to these files do not create a new backup.
5.Retention: Only the five newest versions of a job are retained. On making the sixth one, the oldest version is deleted from the disk.
6.Automatic backups: They occur according to the interval you specify (at least 1 minute). The first automatic backup occurs one interval after the initial setup.

### Restoring a version

1. Choose the job, then the version you want.
2. Choose the folder to restore into. **A new, empty folder is safest.** Files with the same names in that folder are overwritten.
3. Open that folder to see the files as they were at that version.

### Where AutoVault keeps its files

| File or folder | What it holds |
|---|---|
| jobs.dat | Your saved jobs and their version lists. Delete it to start with no jobs |
| AutoVault.log | The activity log: jobs added, backups, restores, retention, and |

| | |
|---|---|
| | errors |
| Backup folder | One sub-folder per version, named `ver` followed by a number. Version names shown in the menu look like `20261031_184123_612` (date and time) |

These files are made in the folder AutoVault runs from.

**Messages you may see**

| Message | Meaning and what to do |
|---|---|
| No changes ... Nothing to do. | Nothing in the source folder has changed since the last backup. This is normal |
| That source folder does not exist. | The path you typed is wrong. Check the spelling, or use **Browse...** in the window |
| Backup folder cannot be inside the source folder. | Choose a backup folder that is outside the source folder |
| The interval must be a whole number. / Interval must be at least 1 minute. | Type a number of 1 or more |
| Backup failed for job ... | The backup could not run, for example because the source folder was deleted. Check the folder and try again |
| Version not found / Job not found | The version or job no longer exists. List them again and pick from the list |
| Please type a number from 1 to N. | Pick one of the numbers shown |

**Tips and limits**

- Because each version is a full copy, backups use disk space. The limit of 5 versions keeps this under control.
- Automatic backups only run while AutoVault is open.
- Restore into a new folder first, and check the files before replacing anything important.
- To see what AutoVault did, open `AutoVault.log` in a text editor.